

✓ CORRELATION & COVARIANCE :

* COVARIANCE

- Covariance describes how two variables change together.
- It doesn't show the strength of the relationship but indicates the direction.
- If both variables increase together $\rightarrow$ +ve relationship
If one increases and other decreases $\rightarrow$ -ve relationship.

VARIABLE 1 $\uparrow$ VAR 2 $\uparrow$ $\Rightarrow$ +ve relationship

VARIABLE 1 $\uparrow$ VAR 2 $\downarrow$ $\Rightarrow$ -ve relationship

- This helps in understanding the nature of relationship between our data points.

* CORRELATION

- Measures the strength and direction of a relationship between 2 variables.
- Positive Values $\rightarrow$ +ve Relationship
Negative Values $\rightarrow$ -ve Relationship
- The co-relation coefficient, represented as r , ranges from -1 to 1. A value of zero indicates NO CORRELATION, whereas values closer to -1 $\rightarrow$ STRONG -ve CORRELATION & closer to +1 $\rightarrow$ STRONG +ve CORRELATION.
- $\therefore -0.5 < r < 0.5 \Rightarrow$ Weak correlation

0.8 $\Rightarrow$ STRONG +ve CORRELATION

→ CALCULATIONS : Two variables X and Y : NUMERICAL

* PEARSON'S CORRELATION COEFFICIENT :-

$$r(x,y) = \frac{\text{cov.}(x,y)}{\text{std}(x) \cdot \text{std}(y)} = \frac{\text{Cov}(x,y)}{\sqrt{\text{var}(x)} \cdot \sqrt{\text{var}(y)}}$$

$$\text{Cov}(x,y) = \sum_{i=1}^n \frac{(x_i - \mu_x)(y_i - \mu_y)}{n}$$

* CAUSATION :

- Causation means change in one variable directly causes a change in another.
- It's a **direct** connection unlike correlation.
- Cause & effect relationship.

* CAUSATION :-

- IF you WATER a plant it grows. Your action of watering directly causes the plant to grow.
- There's a clear cause-and-effect relationship here.

* CORRELATION :-

- Suppose people who own more cars have larger homes.
- This doesn't mean that owning cars causes bigger homes or vice versa.
- It might just mean they're BOTH related to another factor, like income. They move together, but one doesn't necessarily cause the other.

→ In short CAUSATION is DIRECT LINK while correlation is more of an association without direct cause & effect.

* LIMITATIONS OF CORRELATION :-

1. Influenced by OUTLIERS.
2. It does not measure non-linear relationships.

EXAMPLE :- CORRELATION AND COVARIANCE

Q. Suppose you are a financial analyst studying the relationship BETWEEN the stock price of two technology companies, Company A and Company B, over the past years. You have collected daily closing prices for both companies and want to analyze WHETHER THERE is a co-relation or covariance between their stock performance

Company A PRICES : [100, 102, 105, 103, 110, 112, 115, 118, 120, 122]

Company B PRICES : [80, 82, 85, 87, 88, 92, 95, 97, 98, 100]

Solⁿ) Company A Company B

100 80

102 82

105 85

103 87

110 88

112 92

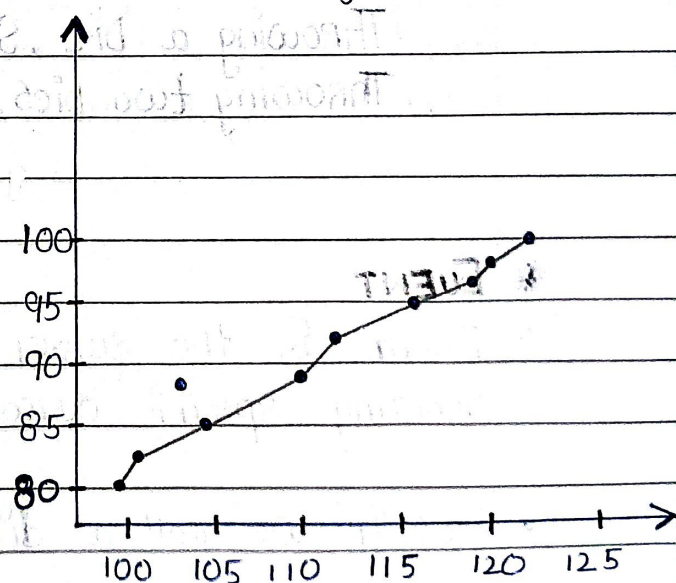
115 95

118 97

120 98

122 100

Company B closing Price



Company A closing Prices

$$r = 0.98$$

$$\text{COV} = \pm 54$$

[+ve CORRELATION]